



152



UCSC

University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester I - 2023

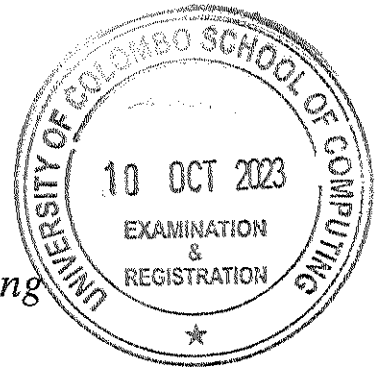
SCS1206 — Laboratory I

Two (2) Hours

Answer All Questions

Number of Pages = 8

Number of Questions = 4



To be completed by the candidate

Index Number

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Important Instructions to candidates

- The students are required to answer all questions within two hours.
- The medium of instructions and questions is English.
- You shall write your answers in **English**.
- This paper has 4 questions on 8 pages.
- Answer **all 4** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Write your index number on each and every page of the question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Calculators are **not allowed**.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). Consider the following directory hierarchy and sequence of commands executed in a Ubuntu GNU/Linux terminal.

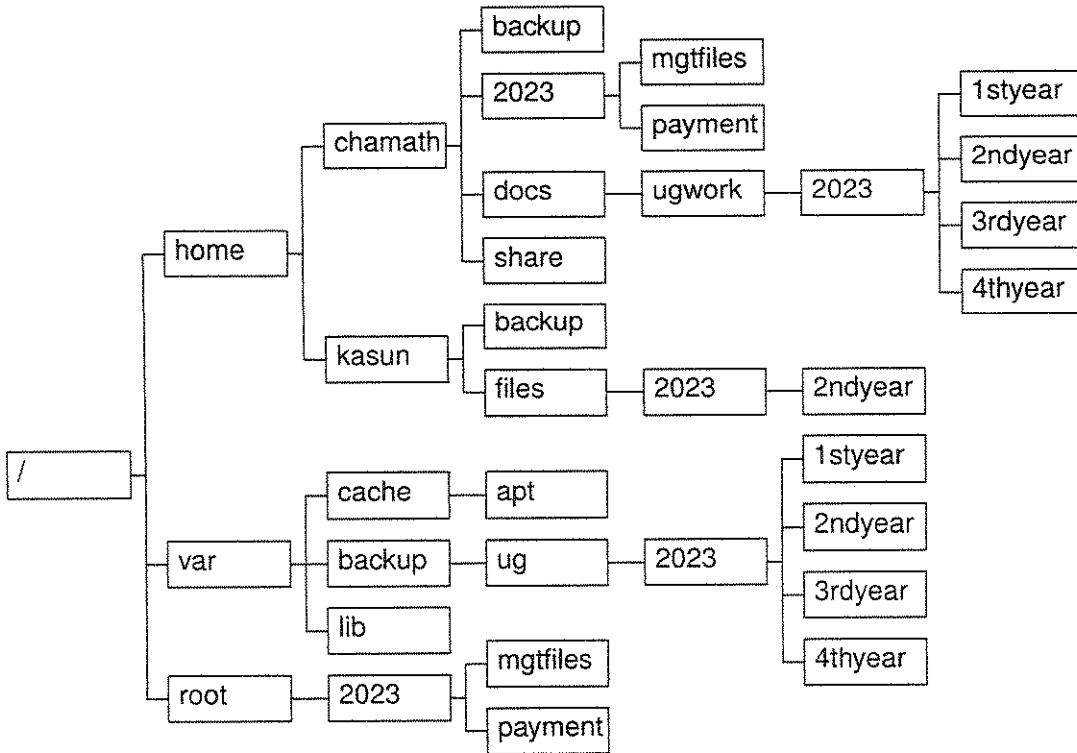


Figure 1: Extract of an Ubuntu GNU/Linux directory structure

Assume you logged in as chamath and executed following commands.

\$ cd docs/ugwork/2023	1
\$ whoami	2
\$ rm 2ndyear	3
\$ ls -a ../	4
\$ mkdir CS/PastPapers	5
\$ cd CS	6
\$ pwd	7
\$ _____	8
# apt-get update	9
# _____	10
# echo "Hello" > text && echo "World" > text	11
# \$?	12
\$ _____	13

Example question and answer:

Q: What is the outcome of the command ① ?

Ans: User chamath change directory to docs/ugwork/2023

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Use the Figure 1 and sequence of commands given to answer the following sub questions.

i. What is the outcome of the command ②?

[1 marks]

ii. What is the output of the command ③?

[2 marks]

iii. What is the outcome of the command ④?

[3 marks]

iv. What is the outcome of the command ⑦?

[2 marks]

v. Write the command chamath has executed as command ⑧?

[2 marks]

vi. Why chamath has executed the command ⑨?

[3 marks]

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vii. chamath wants to install all available software updates for his GNU/Linux system. What command should he use at ⑩?

[2 marks]

viii. Write the content of the file after executing the command ⑪?

[2 marks]

ix. What is the output of the command ⑫?

[1 marks]

x. What is the command chamath should use to access /home/chamath/backup directory using relative path as the command at ⑬?

[2 marks]

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- (b). Assume chamath needs to create an alias `nstat` for the command `netstat -ntlp` and use it whenever he log in as chamath. What is the file chamath should use to create the alias?

[2 marks]

- (c). What is the command chamath should use to get the size of the present working directory?

[3 marks]

2. (a). Given below is a US QWERTY keyboard and answer the sub questions using the knowledge of touch typing.

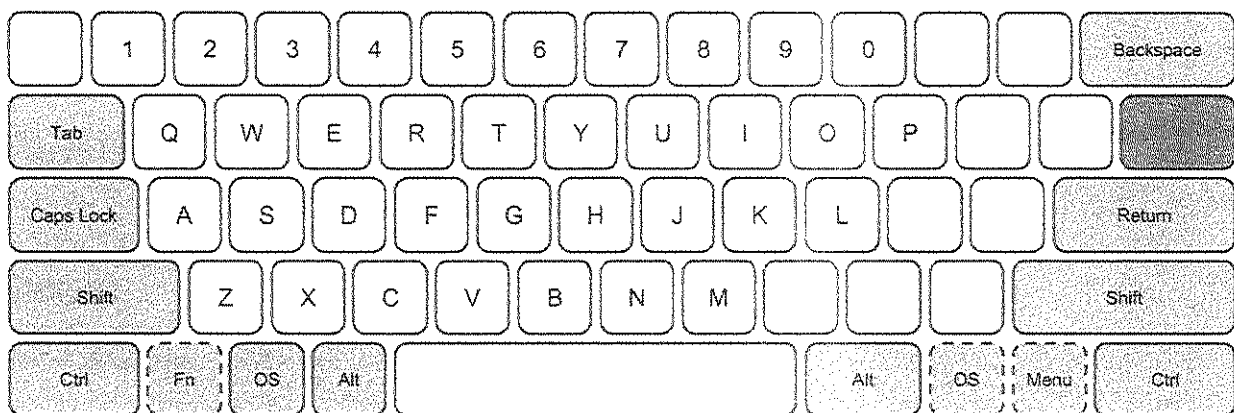


Figure 2: QWERTY Keyboard (credit to Wikipedia, author: Brilliantwiki2)

- i. Write letters of **four (04)** keys typed by using **index finger** of your **right hand**.

[4 marks]

- ii. Write letters of **three (03)** keys typed by using **middle finger** of your **left hand**.

[3 marks]

- (b). Everything in Linux is a file. The result of `ls -l /home/ruwan/documents/` is given below.

```
-rw-r--r-- 1 ruwan staff 97519 Jun 28 2022 confidential.txt
drwxr-x--- 1 ruwan staff 28148 Sep 30 2022 kamal
-rwxr-x--- 1 ruwan staff 67235 May 31 2022 nimal
lrwxr-xr-x 1 ruwan staff 4096 Sep 30 2022 sudath -> kamal
-r--r--r-- 1 ruwan staff 4254912 Aug 19 2022 sign.sh
```

- i. ruwan executed `chmod 650 nimal`. What are the new permissions of the file `nimal`?

[3 marks]

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- ii. ruwan executed `chmod uo+x,g-rw sign.sh`. What are the new permissions of the file `sign.sh`?

[3 marks]

- iii. ruwan want to change the group of the file `sign.sh` to group `kamal`. What is the command ruwan should use?

[2 marks]

- (c). What happens if you execute `!!` in a bash shell?

[2 marks]

- (d). What is the GNU/Linux command that you can use to check the amount of free memory?

[1 marks]

- (e). What is the GNU/Linux command that you can use to get process related information?

[1 marks]

- (f). What is the GNU/Linux command that you can use to get the network configuration information of all network interfaces?

[2 marks]

- (g). Which file of a GNU/Linux system is used to store the user password?

[2 marks]

- (h). Which file of a GNU/Linux system is used to store users group information?

[2 marks]

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3. (a). $\text{\LaTeX}$ is a powerful tool for technical writers.

i. Complete the given $\text{\LaTeX}$ script correctly to generate a valid output.

[14 marks]

```
\documentclass{article}
```

```
-----
```

```
\begin{document}
```

```
\section{Main topic}
```

```
----- % start of environment  
\includegraphics[scale=0.8]{picture}
```

```
----- % name of the picture
```

```
-----
```

```
----- % end of environment
```

```
{ ----- This content is in bold format}
```

```
-----{color of this text is blue}
```

```
Figure \ref{fig:pic} is a picture of Randil
```

```
\end{document}
```

ii. You are required to include a page in landscape orientation at the middle of a document. Describe how you can do it using $\text{\LaTeX}$ commands and parameters.

[6 marks]

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- iii. **Git** is a very popular source code repository version management system created by Linus Torvalds in year 2005. Explain how you can restore a file from a previous commit using required Git commands.

[5 marks]

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4. (a). Write the key or key combinations to achieve the given tasks in a vim editor command mode.

- i. Move the cursor to the end of a line.

[2 marks]

- ii. Delete the current line.

[2 marks]

- iii. Move the cursor 10 words right.

[3 marks]

- iv. Divide the vim editor view horizontally and open a new file named 'Gune'.

[3 marks]

- v. Find and replace the word 'Chair' with 'Sofa'.

[3 marks]

- vi. What is the vim command use to enable the identification of spelling mistakes.

[2 marks]

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(b). List the steps that you take to recover forgotten or lost root password of a GNU/Linux system.

[6 marks]

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(c). List four (04) applications provided by LibreOffice suite.

[4 marks]

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